

Investments in Human Capital in the Age of AI

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Capital Theory Seminar (University of Chicago)

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Roadmap

- 1 Introduction: Past and Future Technological Change
- 2 Model: Optimal Training Choice under Uncertainty
- 3 Measuring the AI Shock
- 4 Calibration
- 5 Simulation of AI impact

Past Technological Change

- Before AI (LLMs): mostly **automation** of “routine” tasks.

Acemoglu and Autor (2011)

- **Highly educated** workers have been **complementary**.

Autor (2019), Acemoglu et al. (2022), Burstein et al. (2019)

- Stronger **incentives** to enroll in **tertiary education**.

AI is Different

- **AI** performs (also) “non-routine” tasks (coding, analysis).
- **Ambiguous impact** on demand for **college workers**:
 - Displacement vs productivity.
- **Not all human capital is the same**:
 - Some training paths face **automation risks**.
 - Others are more likely to be **augmented**.
 - **Uncertainty** not equally distributed.

This Paper

I use a task-based model and US data to answer:

1. **How do individuals adjust human capital investments** in response to AI?
2. Quantifying the impact of AI on **employment and expected wages** for each training.
→ Calibrate the model and simulate AI impact.

Preview of Results

1. Expected automation drives reallocation of labor:

- Fewer students choose training paths exposed to automation
(Computer science, Business, Legal professions).
- Shift towards “safer” paths (Technical Manual jobs, Education)...
→ ... or no training at all.

Preview of Results

1. Expected automation drives reallocation of labor:

- Fewer students choose training paths exposed to automation (Computer science, Business, Legal professions).
- Shift towards “safer” paths (Technical Manual jobs, Education)...
→ ... or no training at all.

2. Labor reallocation partially offsets direct AI impact

- Magnitude depends on students' responsiveness to the AI shock
→ If **high-responsiveness**: expected **wages uniformly up** in all training paths (by 15%).
→ If **low-responsiveness**: less labor reallocation and **differences in wage gains**.

Literature and Contribution

- **AI and the labor market**

Acemoglu and Restrepo (2018); Eloundou et al. (2024); Acemoglu et al. (2025);

Autor and Thompson (2025); Agrawal et al. (2025); Ide and Talamàs (2025)

→ I model **supply-side** response to AI.

- **Human capital under uncertainty**

Levhari and Weiss (1974); Rosen (1986)

→ New source of risk: **training-specific** AI uncertainty.

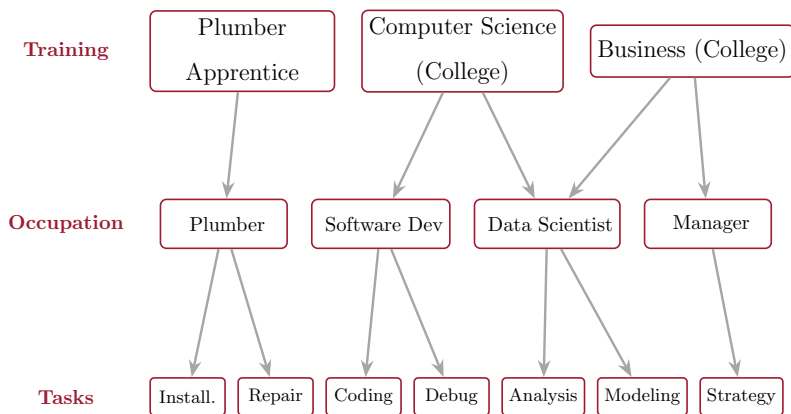
- **Closest paper:** Althoff and Reichardt (2026)

→ **Endogenous training choice**, training-specific risk.

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Training, Occupations, and Tasks



Timeline of the Model

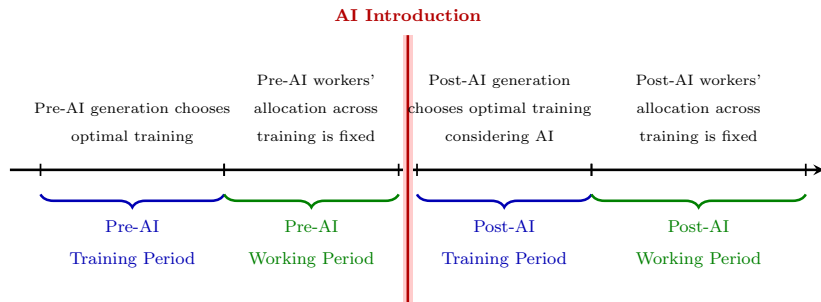


Figure 1: Timeline of the Model.

Working Period

- Each training j linked with bundle of tasks $\tau \in \mathcal{T}_j$.
- Output per worker for training j

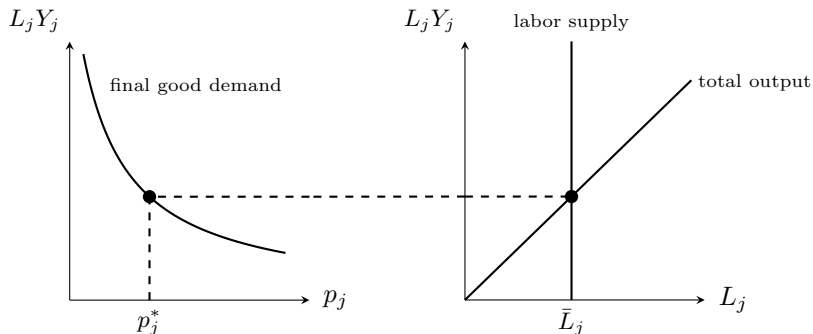
$$Y_j = \left(\sum_{\tau \in \mathcal{T}_j} \underbrace{\theta_{j,\tau}^{\frac{1}{\rho}}}_{\text{task importance}} y_{\tau}^{\frac{\rho-1}{\rho}} \right)^{\frac{\rho}{\rho-1}}$$

$$y_{\tau}(l_{\tau}, k_{\tau}) = \begin{cases} l_{\tau} \gamma_{\tau} & \text{if task is not automatable } (\tau \in \mathcal{N}_j) \\ \phi_{\tau} k_{\tau} & \text{if task is automatable } (\tau \in \mathcal{A}_j) \end{cases}$$

- AI **Automation**: set $\mathcal{A}_j$ expands.
- AI **Augmentation**: higher labor productivity $\uparrow \gamma_{\tau}$

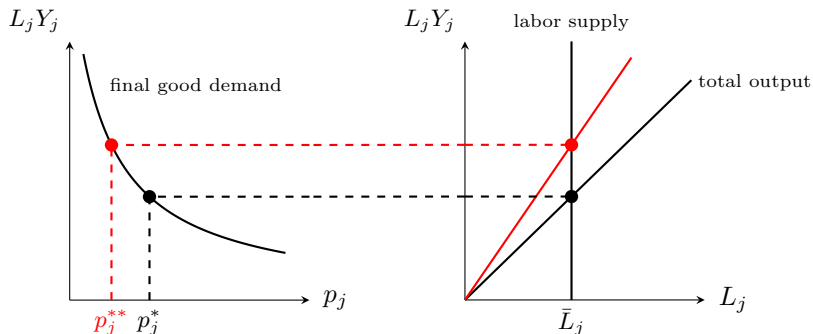
Channels and Mechanisms: Baseline

- Number of workers with training j is fixed at $\bar{L}_j$
- Competitive markets $\implies w_j = p_j Y_j \underbrace{\Gamma_j}_{\text{Labor Share}}$



Channels and Mechanisms: Automation

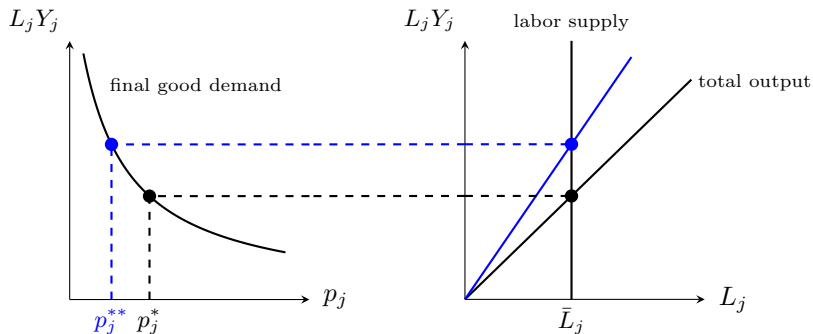
- **Automation:** increase in set of automatable tasks ($\uparrow \mathcal{A}_j$).



Ambiguous impact on wage: $w_j = \downarrow p_j \uparrow Y_j \downarrow \Gamma_j$

Channels and Mechanisms: Augmentation

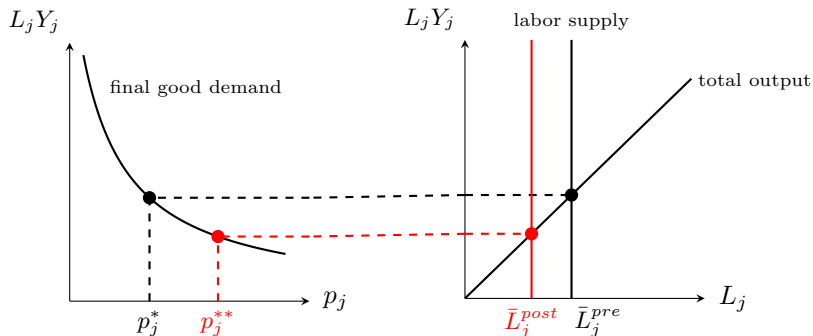
- **Augmentation**: increase in labor productivity ($\uparrow \gamma_\tau$).



Ambiguous impact on wage: $w_j = \downarrow p_j \uparrow Y_j \Gamma_j$

Channels and Mechanisms: Labor Supply Adjustment

- **AI can shift labor supply** for next generation.



Positive impact on wage: $w_j = \uparrow p_j Y_j \Gamma_j$

Training Period

- Agents maximize lifetime utility:

$$\max_j \{ \beta \mathbb{E}_t [u(w_j(\mathcal{A}, \gamma))] - C_{ij} \}$$

- Future automation ($\mathcal{A}$) and augmentation (γ) are unknown.
- Define $a_\tau \in \{0, 1\}$ and $b_\tau \in \{0, 1\}$ as Bernoulli random variables for whether task τ will be **automated**/**augmented** in the future.

$$\underbrace{A_j = \sum_{\tau \in \mathcal{T}_j} \theta_{j,\tau} a_\tau}_{\text{Weighted share of automated tasks}}, \quad \underbrace{\gamma_\tau = 1 + b_\tau (g_\tau - 1)}_{\text{Augmentation factor for task } \tau}$$

From Expected Utility to Certainty Equivalent

- CRRA utility $u(w) = w^{1-\eta}/(1-\eta)$.
- **Separable AI shocks:**
 - Automatable tasks face only **automation** risk.
 - **Augmentation** only in non-automatable tasks.
 - The two shocks are independent.
- **Rational expectations:** agents anticipate price p_j .
 - CES demand across training goods (elasticity ϵ).
- **Log-normal wages:** $\ln w_j$ linearized, assumed normal.
 - Closed form for expected utility.

Solution: Training Choice

- Each training j is summarized by its **Certainty Equivalent Ω_j**

$$\Omega_j = \underbrace{\mathbb{E}(\ln w_j)}_{\text{Expected log-wage}} - \underbrace{\frac{\eta - 1}{2} \text{Var}(\ln w_j)}_{\text{Risk premium}}$$

→ $\mathbb{E}(\ln w_j)$: price, labor share, and productivity. [Details](#)

→ $\text{Var}(\ln w_j)$: uncertainty on **automation** and **augmentation**.

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- Agents solve $\max_j \{\beta \Omega_j - C_{ij}^*\}$, with

$$C_{ij}^* = \underbrace{\bar{C}_j^*}_{\text{average cost}} - \underbrace{\xi_{ij}}_{\sim \text{iid Gumbel}(0, \zeta)}$$

- The share of labor choosing training j is

$$L_j^* = \frac{\exp((\beta \Omega_j - \bar{C}_j^*)/\zeta)}{\sum_{h=0}^J \exp((\beta \Omega_h - \bar{C}_h^*)/\zeta)}$$

Roadmap

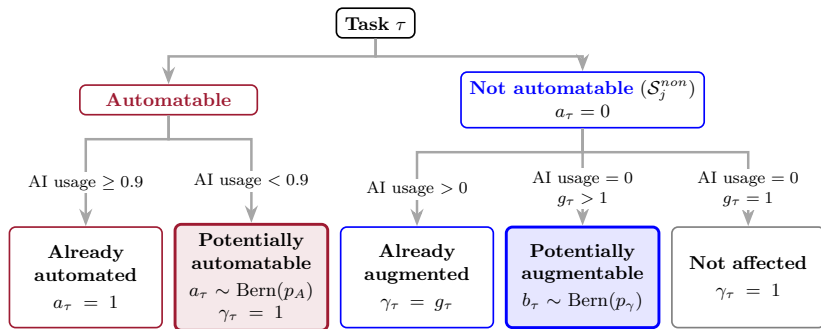
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Task-level AI measures

Three measures at the **task level**:

- **Automatable** $\in \{0, 1\}$: AI can perform the entire task. Althoff and Reichardt (2026) $\rightarrow \text{Automatable} = 0 \implies a_\tau = 0$
- **Augmentation** ($g_\tau \geq 1$): potential human productivity boost (1.1 = 10% productivity boost). Althoff and Reichardt (2026)
- **AI penetration**: current AI usage. Massenkoff and McCrory (2026) $\rightarrow \in [0, 1]$: closer to 1 means more automation.

Classifying tasks: Automation vs Augmentation



- Uncertainty only in **potential** tasks (shaded boxes).

Expected Automation and Augmentation

- Expected Automation** $\mu_{A,j} = \underbrace{S_{A,j}^{cur}}_{\text{already autom.}} + p_A \underbrace{S_{A,j}^{pot}}_{\text{pot. autom.}}$

- Expected Augmentation Gain** (only non-automatable tasks)

$$\sum_{\text{already augm.}} \theta_{j,\tau} d_\tau + p_\gamma \sum_{\text{pot. augm.}} \theta_{j,\tau} d_\tau, \quad \underbrace{d_\tau = 1 - g_\tau^{\rho-1}}_{\text{Task productivity gain}}$$

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- **Uncertainty** $\propto p(1-p)$, and it is higher when
 - the share of **potential** tasks is larger.
 - potential tasks are **concentrated** in few tasks.
- **Baseline:** $p_A = p_\gamma = 0.5 \implies$ agnostic workers.

Automation by Training Path Alternative scenario

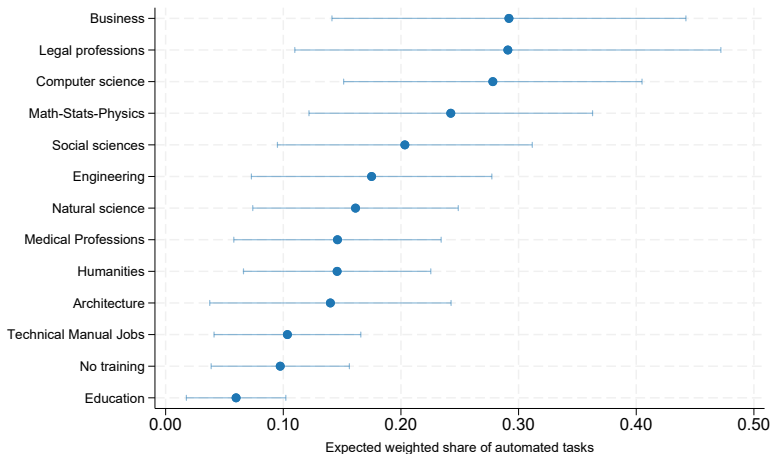


Figure 2: Expected Automation across Training Paths (with 1 sd CI).

Augmentation by Training Path

Alternative scenario

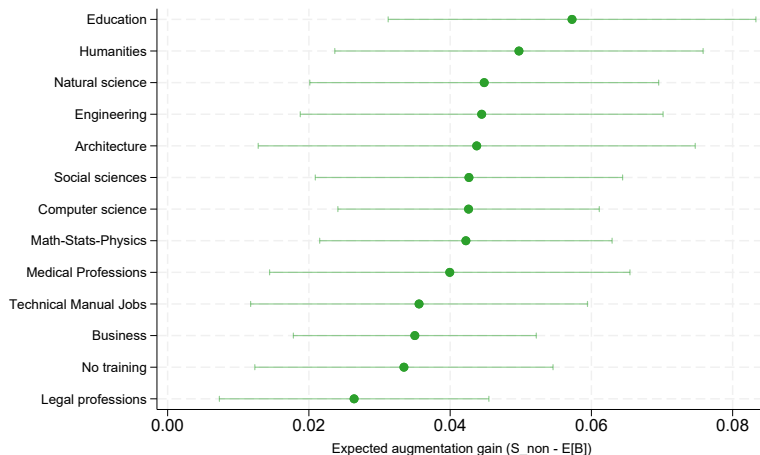


Figure 3: Expected Augmentation gain across Training Paths (1 sd CI).

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13 training paths: 11 college fields + technical manual + no training.

- **O*NET:** tasks and their importance in each occupation.
- **CIP-SOC crosswalk:** maps fields of degree to occupations.
- **ACS 2022** (employed, older than 30):
 - employment shares L_j^{2022} .
- **BLS 2022:**
 - median wages w_j^{2022} .
 - on-the-job training: technical manual jobs vs no training.

Calibration: Pre-AI Equilibrium (2022)

- **No AI in 2022:** no automation, augmentation, or uncertainty.
→ $\Omega_j^{pre} = \ln w_j^{2022}$.

- **Training costs:** invert the logit (no training: $\bar{C}_0^* = 0$)

$$\bar{C}_j^* = \beta (\ln w_j^{2022} - \ln w_0^{2022}) - \zeta (\ln L_j^{2022} - \ln L_0^{2022})$$

→ High cost: high wage premium, but few choose j .

- **Demand weights α_j :** from 2022 wages and employment.
- I follow the literature for the other parameters

Other Parameters

Responsiveness of Training Choices

- ζ governs how strongly students respond to wages:

$$e_j \equiv \frac{\partial \ln L_j^*}{\partial \Omega_j} = \frac{\beta(1 - L_j^*)}{\zeta} \approx \frac{\beta}{\zeta}$$

- Estimates of e differ by more than an order of magnitude
→ three scenarios:

Scenario	ζ	e	Evidence
High (benchmark)	0.053	≈ 15	Task-based model (Althoff and Reichardt, 2026)
Medium	0.5	≈ 1.6	Occupation choice (Burstein et al., 2019)
Low	1.2	≈ 0.67	College majors (Long et al., 2015)

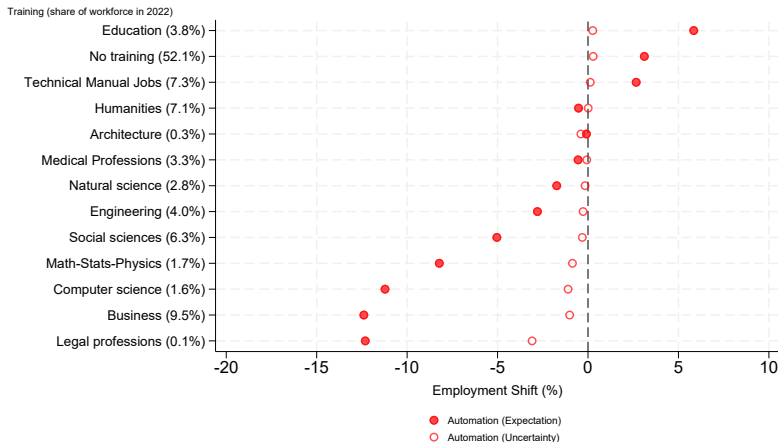
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Labor Reallocation (high-response): Automation

Expected Automation: no augmentation and no uncertainty.

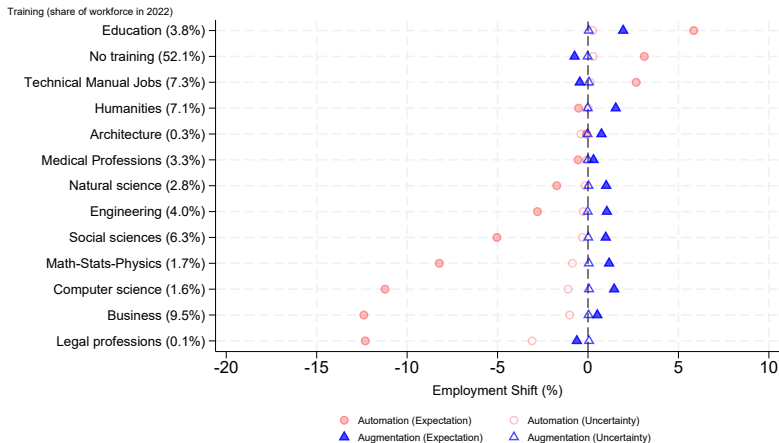
Automation Uncertainty: labor reallocation due to risk aversion.



Labor Reallocation (high-response): Augmentation

Expected Augmentation: no automation and no uncertainty.

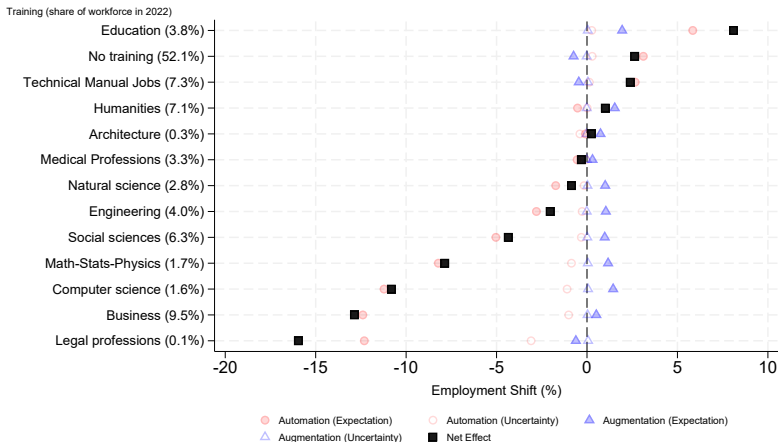
Augmentation Uncertainty: labor reallocation due to risk aversion



Labor Reallocation (high-response): Net Effect

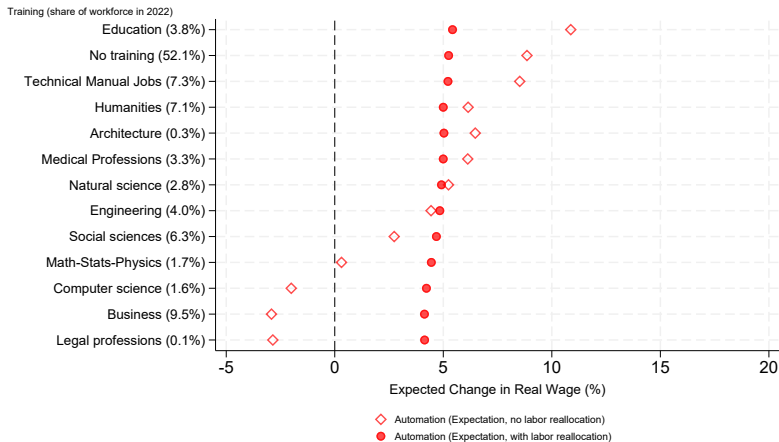
Net effect: Final impact on employment.

Sensitivity



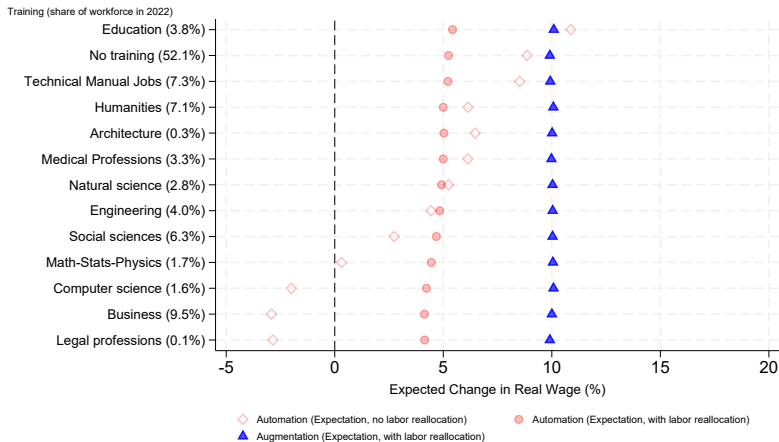
Expected wages (high-response): Automation

Expected Automation: the impact of labor reallocation on wages.



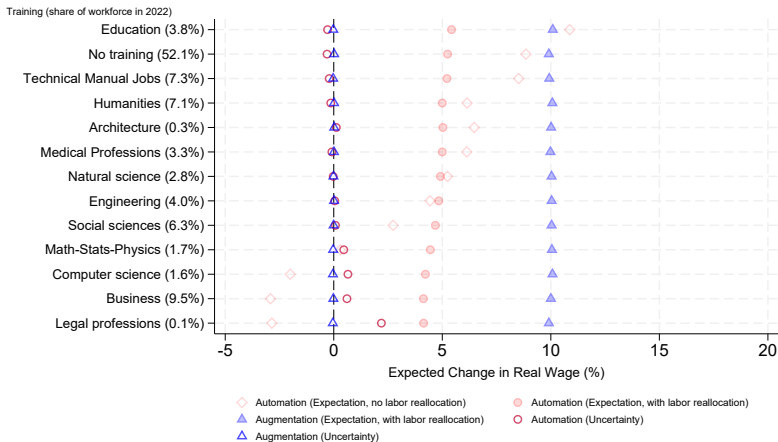
Expected wages (high-response): Augmentation

Expected Augmentation: the impact of augmentation on wages.



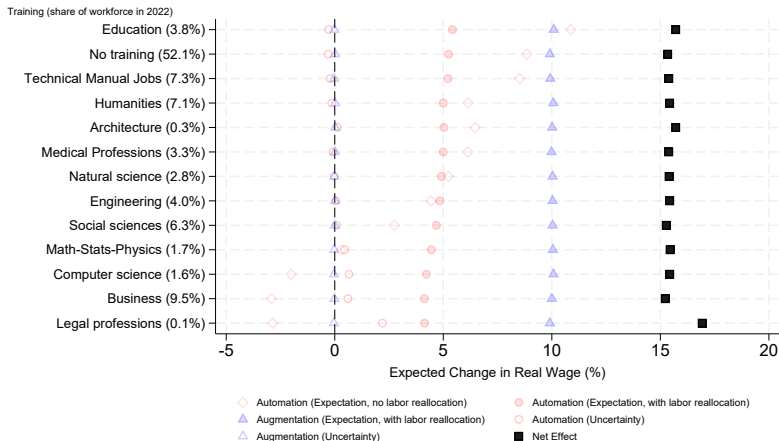
Expected wages (high-response): Uncertainty

Uncertainty: risk premium due to uncertainty.



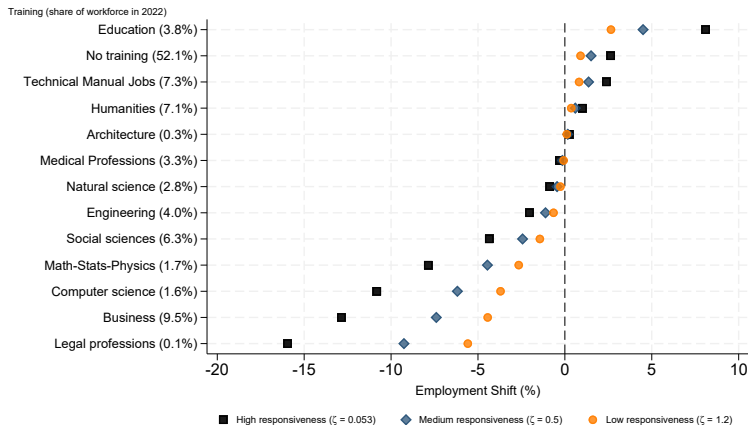
Expected wages (high-response): Net Effect

Net effect: Final impact on expected wages. Sensitivity



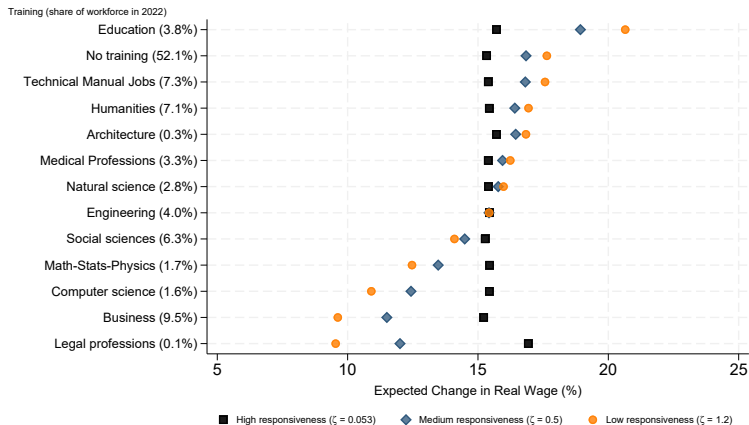
Responsiveness of Training Choices: Employment

Same ranking in all scenarios, but **magnitude** scales with e .



Responsiveness of Training Choices: Expected Wages

Average gain $\approx +15\%$; lower $e \implies$ wage gaps open.



Conclusion

- AI has the potential to shift human capital allocation.
- Most of the labor reallocation is driven by expected automation.
- Labor scarcity partially offsets the direct impact of AI on wages.
- Magnitude depends on students' responsiveness.

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Automation and Augmentation Uncertainty

$\kappa \in [0, 1]$: correlation of AI success across tasks within a training path.

→ $\kappa = 0$: tasks are independent.

→ $\kappa = 1$: AI succeeds on all potential tasks or on none.

$$\sigma_{A,j}^2 = p_A(1 - p_A) \left[(1 - \kappa) \underbrace{\sum_{\text{pot. autom.}} \theta_{j,\tau}^2}_{\text{Concentration (HHI)}} + \kappa \underbrace{\left(S_{A,j}^{\text{pot}} \right)^2}_{\text{Size of the gap}} \right]$$

$$\sigma_{B,j}^2 = p_\gamma(1 - p_\gamma) \left[(1 - \kappa) \sum_{\text{pot. augm.}} \theta_{j,\tau}^2 d_\tau^2 + \kappa \left(\sum_{\text{pot. augm.}} \theta_{j,\tau} d_\tau \right)^2 \right]$$

with $d_\tau = 1 - g_\tau^{\rho-1}$ (task productivity gain). Baseline: $\kappa = 0.5$.

Expected Log-Wage [Back](#)

$$\mathbb{E}(\ln w_j) = \underbrace{\ln p_j}_{\text{Price}} + \underbrace{\frac{1}{1-\rho} \ln \Gamma_j(\mu_{A,j})}_{\text{Labor share}} + \underbrace{\frac{1}{\rho-1} \ln \mu_{\Pi,j}}_{\text{Productivity}}$$

- $\mu_{\Pi,j} = \mathbb{E} \left[\sum_{\tau \in \mathcal{T}_j} \theta_{j,\tau} (1 - a_\tau) \gamma_\tau^{\rho-1} \right]$: expected productivity index of the tasks still performed by humans.
- p_j is an equilibrium object: it depends on the reallocation of labor across all training paths.

Automation by Training Path (Alternative) [Back](#)

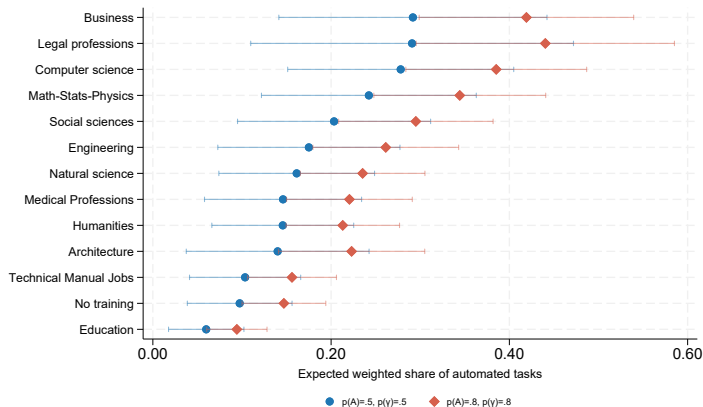


Figure 4: Expected Automation across Training Paths: $p = 0.5$ vs $p = 0.8$ (with 1 standard deviation confidence interval).

Augmentation by Training Path (Alternative)

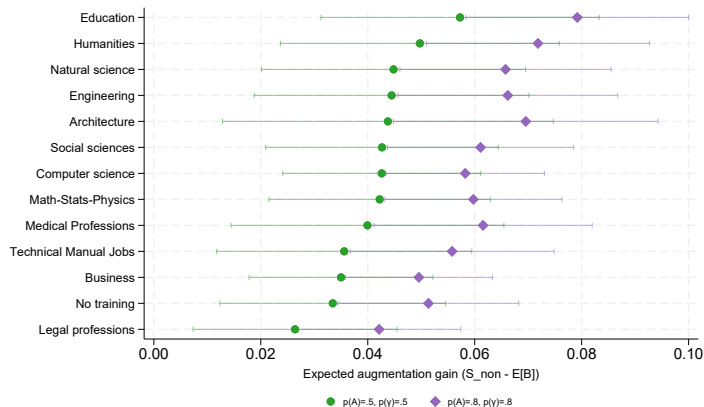
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Figure 5: Expected Augmentation across Training Paths: $p = 0.5$ vs $p = 0.8$ (with 1 standard deviation confidence interval).

Calibrated Parameters [Back](#)

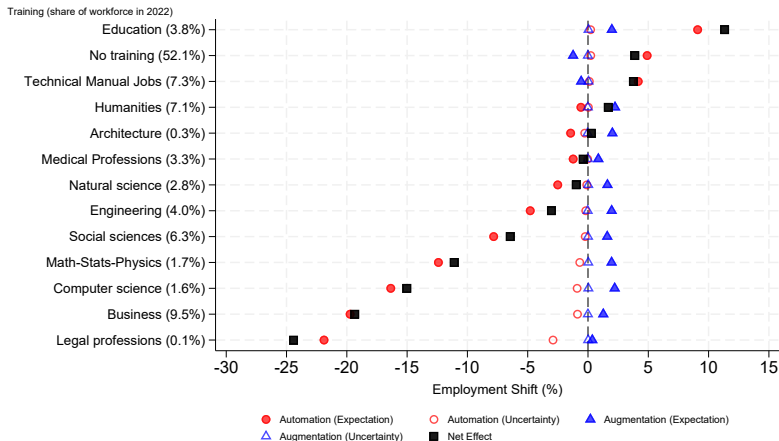
	Description	Value	Source
ρ	Substitution across tasks	0.49	Humlum (2021)
ϵ	Substitution across goods	1.57	Burstein et al. (2019), Caunedo et al. (2023)
χ	Relative cost of AI capital	0.73	Acemoglu (2025)
ζ	Responsiveness of training	0.053 / 0.5 / 1.2	Three scenarios
β	Discount factor (7 years)	0.813	3% annual rate
η	Relative risk aversion	5	Sensitivity: 2, 10
p_A, p_γ	Prob. of automation / augm.	0.5	Sensitivity: 0.8
κ	Within-path task correlation	0.5	Midpoint of [0, 1]
$\bar{C}_j^*, \alpha_j$	Training costs, demand weights	Calibrated	2022 employment, wages

$$\text{Labor share: } \Gamma_j = \frac{\chi^{\rho-1} (1 - \mu_{A,j})}{\mu_{A,j} + \chi^{\rho-1} (1 - \mu_{A,j})} \quad (= 1 \text{ before AI})$$

AI impact on employment: Alternative Scenario

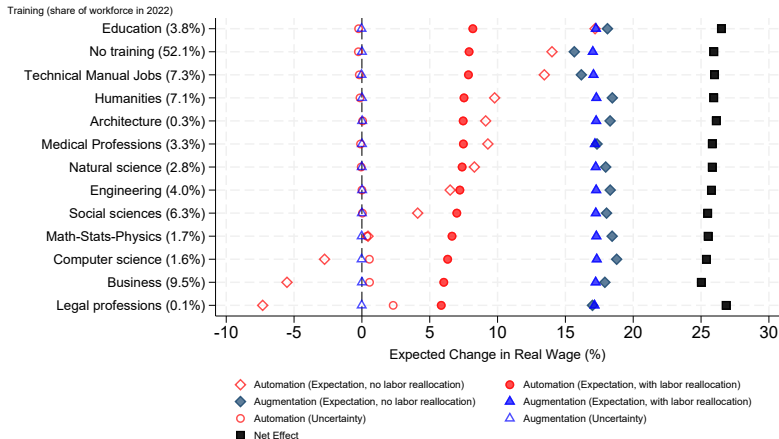
[back](#)

Alternative scenario: $p_A = p_\gamma = 0.8$



AI impact on wages: Alternative Scenario [back](#)

Alternative scenario: $p_A = p_\gamma = 0.8$



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